

Chapter-4 – Spreadsheet (LibreOffice-Calc)

- A spreadsheet is a digital application or software tool used to organize, analyze, and store data in a tabular form
- LibreOffice is a free and open-source spreadsheet program, part of the LibreOffice suite.

Basic Components

- **Spreadsheet:**

Collection of sheets, each containing rows and columns.

- **Sheet:**

A single page in a spreadsheet file. Multiple sheets can exist in one file.

- **Cells:**

Intersection of a row and column (e.g., A1, B2).

- **Rows:**

Horizontal arrangement of cells

Rows are labelled as **1, 2, 3...**

Total no of rows- 10,48,576

Shortcut to go last row-

CTRL + Down arrow

By default height – 0.45 cm

- **Columns:**

Vertical arrangement of cells

Column are labelled as **A, B, C...**

Total no of Columns – **16384 (A to XFD)**

*In older version 7.3 and below it was –

1024 (A to AMJ)

Shortcut to go last Column –

CTRL + Left arrow

By default width- 2.26 cm

- **Formula Bar:**

A section at the top where you can view or enter data or formulas for a selected cell.

Formula bar is Automatically Activated when we press **equals to symbol (=)**

- **Name box** : On left hand side of Formula bar name box is situated it show address of active cell.
- **Cell pointer** : A highlighted border around the currently active cell.

Navigation and Selection Techniques

- **Moving Between Cells:**
 1. **Arrow keys**: Move one cell at a time.
 2. **CTRL + Arrow keys**: Jump to the edge of data.
(LEFT , RIGHT,TOP, BOTTOM)
- **Shortcut for Specific Locations:**
 1. **CTRL + Home**: Jump to the first cell (A1).
 2. **CTRL + End**: Jump to the last used cell.
- **Selecting Multiple Cells:**
 1. **Shift + Arrow keys**: Select a range of cells.
 2. **CTRL + A**: Select the entire sheet.

Basic Formulas and Functions

What is a Formula?:

A formula performs a calculation (e.g., =A1+B1).

What is a Function?:

A predefined operation (e.g., =SUM(A1:A10)).

Types of Functions-

- 1.Math's functions**
- 2.Statistical function**
- 3.Text function**
- 4.Logical functionDate and time functions**

Maths functions

<u>Function</u>	<u>Work</u>	<u>Syntax</u>	<u>Example</u>
SUM	Adds all the numbers in a range of cells.	SUM(range)	SUM(A1:A5)
AVERAGE	Returns the average of the numbers.	AVERAGE(range)	AVERAGE(A1:A5)
ROUND	Rounds a number to a specified number of digits.	ROUND(number, digits)	ROUND(12.345, 2)
INT	Rounds a number down to the nearest integer.	INT(number)	INT(12.7)
CEILING	Rounds a number up to the nearest multiple of significance.	CEILING(number, significance)	CEILING(12.3, 1)
FLOOR	Rounds a number down to the nearest multiple of significance.	FLOOR(number, significance)	FLOOR(12.7, 1)
MOD	Returns the remainder after a number is divided by a divisor.	MOD(number, divisor)	MOD(10, 3)
POWER	Returns the result of a number raised to a power.	POWER(base, exponent)	POWER(2, 3)
SQRT	Returns the square root of a number.	SQRT(number)	SQRT(16)

EXP	Returns e raised to the power of a given number.	EXP (number)	EXP (1)
LOG	Returns the logarithm of a number to a specified base.	LOG (number, base)	LOG (100, 10)
PI	Returns the value of Pi.	PI ()	PI ()
RANDBETWEEN	Returns a random integer between two numbers.	RANDBETWEEN (lower, upper)	RANDBETWEEN (1, 100)

LOGICAL FUNCTION

Function	Description	Syntax	Example
AND	Returns TRUE if all arguments are TRUE.	AND(condition1, condition2, ...)	AND(A1>0, B1<10)
OR	Returns TRUE if at least one argument is TRUE.	OR(condition1, condition2, ...)	OR(A1>10, B1<5)
NOT	Reverses the logical value of its argument.	NOT(condition)	NOT(A1=5)
IF	Returns one value if a condition is TRUE and another if FALSE.	IF(condition, value_if_true, value_if_false)	IF(A1>10, "Yes", "No")

STATISTICAL FUNCTION

Function	Description	Syntax	Example
COUNT	Counts the number of numeric values in a range.	COUNT(range)	COUNT(A1:A10)
COUNTA	Counts the number of non-empty cells in a range.	COUNTA(range)	COUNTA(A1:A10)
COUNTBLANK	Counts the number of empty cells in a range.	COUNTBLANK(range)	COUNTBLANK(A1:A10)
COUNTIF	Counts the number of cells that meet a specific condition.	COUNTIF(range, condition)	COUNTIF(A1:A10, ">5")
AVERAGE	Returns the average (arithmetic mean) of the numbers in a range.	AVERAGE(range)	AVERAGE(A1:A10)
AVERAGEIF	Returns the average of cells that meet a specific condition.	AVERAGEIF(range, condition)	AVERAGEIF(A1:A10, ">5")
MEDIAN	Returns the median of a group of numbers.	MEDIAN(range)	MEDIAN(A1:A10)

MODE	Returns the most frequently occurring value in a range.	MODE(range)	MODE(A1:A10)
MAX	Returns the largest number in a range.	MAX(range)	MAX(A1:A10)
MIN	Returns the smallest number in a range.	MIN(range)	MIN(A1:A10)

Text Function

Function	Description	Syntax	Example
LEFT	Returns a specified number of characters from the start of a text string.	LEFT(text, num_chars)	LEFT("LibreOffice", 5) → Libre
RIGHT	Returns a specified number of characters from the end of a text string.	RIGHT(text, num_chars)	RIGHT("LibreOffice", 6) → Office
MID	Returns a portion of a text string starting at a specified position.	MID(text, start, num_chars)	MID("LibreOffice", 6, 4) → eoff
LEN	Returns the number of characters in a text string.	LEN(text)	LEN("LibreOffice") → 11
TRIM	Removes all extra spaces from text except for single spaces between words.	TRIM(text)	TRIM(" Hello World ") → Hello World

CONCATENATE	Combines multiple text strings into one.	CONCATENATE(text1, text2, ...)	CONCATENATE("Hello", " ", "World") → Hello World
SEARCH	Finds the position of a text string within another text string (case-insensitive).	SEARCH(find_text, within_text, [start])	SEARCH("Office", "LibreOffice") → 6
FIND	Finds the position of a text string within another text string (case-sensitive).	FIND(find_text, within_text, [start])	FIND("O", "LibreOffice") → 6
UPPER	Converts text to uppercase.	UPPER(text)	UPPER("LibreOffice") → LIBREOFFICE
LOWER	Converts text to lowercase.	LOWER(text)	LOWER("LibreOffice") → libreoffice
PROPER	Capitalizes the first letter of each word.	PROPER(text)	PROPER("libre office") → Libre Office
SUBSTITUTE	Replaces occurrences of a text string with another string.	SUBSTITUTE(text, old_text, new_text, [instance])	SUBSTITUTE("Apple", "p", "b") → Abble
REPLACE	Replaces part of a text string with another string, based on position.	REPLACE(text, start, num_chars, new_text)	REPLACE("LibreOffice", 6, 6, "Calc") → LibreCalc
TEXT	Formats a number and converts it to text.	TEXT(value, format)	TEXT(1234.56, "\$#,##0.00") → \$1,234.56
VALUE	Converts a text string into a number.	VALUE(text)	VALUE("123") → 123
CHAR	Returns the character specified by a number (ASCII).	CHAR(number)	CHAR(65) → A
REPT	Repeats a text string a specified number of times.	REPT(text, number)	REPT("Libre", 3) → LibreLibreLibre

Date and Time function

Function	Description	Syntax	Example
TODAY	Returns the current date.	TODAY()	TODAY() → 2024-12-20
NOW	Returns the current date and time.	NOW()	NOW() → 2024-12-20 10:30:45
DATE	Returns the date for a given year, month, and day.	DATE(year, month, day)	DATE(2024, 12, 25) → 2024-12-25
TIME	Returns the time for a given hour, minute, and second.	TIME(hour, minute, second)	TIME(10, 30, 0) → 10:30:00
YEAR	Returns the year from a date.	YEAR(date)	YEAR(DATE(2024, 12, 25)) → 2024
MONTH	Returns the month from a date.	MONTH(date)	MONTH(DATE(2024, 12, 25)) → 12
DAY	Returns the day of the month from a date.	DAY(date)	DAY(DATE(2024, 12, 25)) → 25
HOUR	Returns the hour from a time.	HOUR(time)	HOUR(TIME(10, 30, 0)) → 10
MINUTE	Returns the minute from a time.	MINUTE(time)	MINUTE(TIME(10, 30, 0)) → 30
SECOND	Returns the second from a time.	SECOND(time)	SECOND(TIME(10, 30, 45)) → 45
WEEKDAY	Returns the day of the week as a number (1 = Sunday by default).	WEEKDAY(date, [type])	WEEKDAY(DATE(2024, 12, 20)) → 6
DAYS	Returns the number of days between two dates.	DAYS(end_date, start_date)	DAYS(DATE(2024, 12, 31), DATE(2024, 12, 20)) → 11
NETWORKDAYS	Returns the number of working days between two dates.	NETWORKDAYS(start_date, end_date, [holidays])	NETWORKDAYS(DATE(2024, 12, 20), DATE(2024, 12, 31)) → 8

Types of Cell Referencing

Each cell in a worksheet has a unique address that is used to refer to it. When we specify a cell address in a formula, LibreOffice Calc reads the value of that cell address. In LibreOffice Calc, a cell address can be specified in the following three ways:

1. Relative Cell Referencing
2. Absolute Cell Referencing
3. Mixed Cell Referencing

1. Relative Cell Referencing

When we copy a formula from one cell to another, LibreOffice Calc automatically adjusts the references for each cell in the formula. In this type of cell address, the column letter comes first, followed by the row number, such as A1, B5, H3, etc.

2. Absolute Cell Referencing

In absolute referencing, if a formula is copied from one cell to another, the cell address reference in the formula does not change.

To make a cell address absolute, a dollar sign (\$) is added before both the column letter and the row number. For example, \$A\$1, \$B\$5, \$H\$3, etc.

3. Mixed Cell Referencing

Each cell address consists of two parts: a column letter and a row number.

If one part of a cell address is relative and the other part is absolute, the cell address becomes mixed. Examples of mixed cell referencing include \$A1, A\$1, \$B3, B\$3, \$H3, and H\$3.

Note -we can use F4 shortcut for cycle referencing

Types of Referencing error

1. #VALUE!

- **Cause:** This error occurs when a formula uses an incorrect data type.
- **Example:**

Adding text and numbers: =A1 + "text"

2. #NUM!

- **Cause:** This error happens due to an invalid numeric operation.
- **Examples:**

Square root of a negative number: =SQRT(-1)

3. #DIV/0!

- **Cause:** Division by zero or an empty cell used as a divisor.

- **Example:**

=A1 / B1 where B1 is 0 or empty.

4. #REF!

- **Cause:** Invalid cell reference due to deleted or moved cells.

- **Example:**

=A1+B1 where B1 has been deleted.

5. #NAME?

- **Cause:** Unrecognized function name, range name, or text string.

- **Examples:**

Typo in a function: =SUME(A1:A10) instead of =SUM(A1:A10).

6. #N/A

- **Cause:** No valid result is available for the formula.

- **Example:**

Lookup functions like VLOOKUP can't find the specified value.

7. Err:502

- **Cause:** Invalid argument or incorrect function usage.

- **Example:**

Improper range in a function: =SUM(A1:A2 B3) (missing

